Lethe

The private pool for frontier AI.

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Lethe is a shared anonymity pool in front of GPT, Claude and Gemini. Firms and people send their AI work through it, and the providers see a crowd instead of a customer. Every member makes every other member harder to identify, so the pool gets stronger as it grows. Firms pay for membership, because for a fund or a bank the question itself is the secret. What you write still reaches the model; Lethe hides who wrote it, and measures what the text itself gives away.

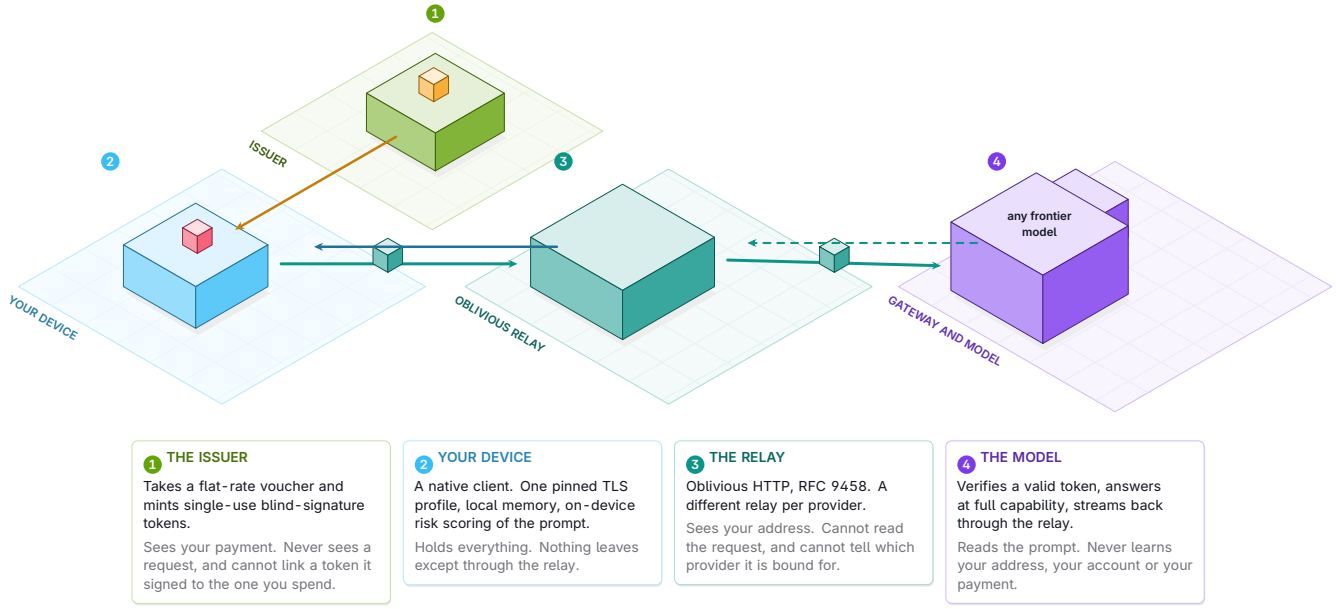


Figure 1. Four parties, and no single one of them can both identify you and read you. The issuer sees your money and never a request. The relay sees your address and never your content. The model reads your prompt and never learns who sent it. This is the same split Apple and Meta already run in production for their own models.

01 The problem changed this year

Three things became true in 2026 that were not true when earlier attempts at this failed.

The profile became a revenue line. OpenAI shipped advertising in February on US free and Go tiers, targeted on conversation topic, past chats and prior ad interactions [1]. In June it published *Dreaming*, background consolidation that builds narrative prose dossiers about a user between sessions [2]. Its own EU policy concedes that using memory for personalised advertising requires affirmative consent, while non-consented contextual ads are limited to the live conversation. The industry has already stated what memory is for. This is no longer a retention policy that a contract can fix. It is a business model, and it will get better at its job rather than worse.

The cross-provider join stopped being hypothetical. Figure 2. Measured on 18 September 2026, seven major endpoints all answer with Cloudflare headers: OpenAI, Anthropic, Mistral, xAI, Together, Groq and OpenRouter. Anthropic's endpoint resolves into Anthropic's own address block and still answers with those headers, so DNS and WHOIS give the wrong answer. Google is the only major provider fronting itself. One commercial party terminates TLS for at least seven providers, which means joining one person's activity across all of them requires no data broker and no subpoena, only a query against one edge network's logs.

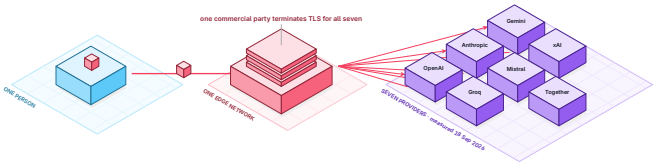


Figure 2. The join needs no broker. One person, seven providers, one edge network in between. Measured directly, not inferred.

A cruder join is already running. Koi Security found eight browser extensions from a single publisher, more than eight million installs, capturing prompts and responses from ten AI platforms including ChatGPT, Claude, Gemini, Copilot and Perplexity before they render, with no opt-out, added silently in a July 2025 update, flowing to a company affiliated with a data broker [3]. That is why we ship a native client and not a browser extension.

Meanwhile the tools to stop it finished. Privacy Pass became RFC 9576, 9577 and 9578 in June 2024, over the VOPRF of RFC 9497 and blind RSA of RFC 9474. Oblivious HTTP is RFC 9458 and is deployed by Apple, Google, Mozilla and Meta. Apple Private Cloud Compute and Meta's WhatsApp Private Processing both ship anonymous credentials, oblivious transport and attested compute for language-model workloads, in production, at hundreds of millions of

users [4]. Kagi shipped blind-signature tokens for paid search in February 2025 and can no longer link a query to a subscriber [5]. Every primitive exists. Nobody has assembled them in front of a frontier model.

02 What we are making

What is your company going to make?

Lethe is a private pool for frontier AI, a dark pool for model queries. It is not a wrapper. A wrapper puts a screen on someone else's model, is no better with a million users than with ten, and can be rebuilt in a weekend. A pool is the opposite: its value rises with every member, because anonymity loves company [14]; no customer can build it alone, because a firm running its own proxy is still the one firm using that proxy; and the provider cannot build it, because hiding a customer from yourself is a promise against yourself.

There are two doors into the pool. **Firms**, funds, banks, law firms, pharmaceutical research and corporate development, send the work they cannot put their name on: deal analysis, research, strategy. Their compliance team keeps a complete record and the provider keeps nothing that points to them. **People** get one native app for every frontier model, with a memory that follows them from model to model and never leaves their machine. Every member, firm or person, uses the same tokens, relays and client fingerprint, so no provider can tell a hedge fund's request from a student's.

A request leaves your device carrying a single-use blind-signature token instead of an account. It crosses an oblivious relay that can see your address but not your content, and a different relay is used per provider, so no intermediary accumulates a picture of where you go. The provider verifies the token is valid, answers at full capability, and streams back along the same path. Your membership was bought from an issuer that signs tokens without seeing them, so it cannot link the token it signed to the one you spent.

Before anything leaves, an on-device pass scores what the text itself still carries and shows you. Identifiers can be replaced with realistic surrogates and restored locally on the way back. We report a residual risk rather than claiming zero, because the model has to read the prompt.

03 How it works, in order

Figure 3. The build order follows the cost of each channel, cheapest first.

Tier one, cryptographic, weeks each against published specifications. Blind-signature access tokens; Kagi's implementation is the template at 216 bytes per single-use token and about a second to mint five hundred. Oblivious HTTP with a third-party relay, taking care never to choose a relay that also fronts the upstream provider. Flat-rate prepaid vouchers through a merchant of record, which sells time rather than metered usage and so avoids both the money-transmitter question and the unstandardised anonymous-metering cryptography. Constant-rate response padding, which is mandatory rather than optional because the token-length side channel reconstructs 27 % of streamed responses and the topic of 53 % [6], and Oblivious HTTP explicitly does not cover traffic analysis. And a native client with one pinned TLS profile.

Tier two, real cost and real benefit. A local quasi-identifier pass using adversary-in-the-loop rewriting rather than a fixed list of PII types, spent against a per-session risk budget instead of applied blanket. Timing decorrelation. Memory held locally by default.

Tier three, optional, high-risk sessions only. Style obfuscation. It has the worst privacy per unit of cost on the list: it damages the text, loses most of its effect against a retrained attacker, and is itself detectable, which makes "this user obfuscates" a durable label.

> **WHAT WE DO NOT BUILD**
> Mixnets, which cap near 100 KB/s with tail latency around three seconds. Homomorphic or multi-party inference for proprietary weights, which is three to four orders of magnitude off and works only on open weights. Word-level differential privacy rewriting, which is reconstructed at 72 to 94 % at the epsilons where utility survives. And we do not lean on confidential compute for the trust story: independent 2026 measurements put overhead between 1 and 118 %, not the single digits vendors quote, and attestation has been forged with $47.62 of parts [7].

04 Why this is defensible

Five reasons, in descending order of durability.

The platforms structurally cannot ship it. A frontier lab can add provider-side redaction next quarter and probably will. It cannot ship cross-provider unlinkability, because that is a promise about what a competitor must not be able to correlate. It cannot ship a log proving what it did not receive, because a claim by the observed party about the observer is not evidence. Both of our core guarantees are assertions against the provider, so they can only be issued by something the provider does not control.

There is a number for what "cannot" is worth. Proton complied with 88.4 % of legal orders against Mail between 2017 and 2025, and denied 410 of 410 against its VPN, because it is unable to comply even when the order binds [8]. That gap between *will not* and *cannot* is the entire product line, and it is measured rather than asserted.

No utility cost on the channels we lead with. The privacy-versus-capability tradeoff, which is the central weakness of every redaction product, simply does not apply to the account, the payment, the network path or the cross-provider join.

No dependence on detector recall. A redaction product's guarantee is only as strong as its weakest named-entity pass. Ours does not rest there.

The patent white space sits exactly here. No filing claims per-recipient unlinkability for *model providers*, negative-disclosure attestation, or detokenisation bound to a tool's action boundary (§11).

05 Why us, and how we know

Why did you pick this idea? Do you have domain expertise? How do you know people need it?

[FILL: the origin story. The strongest version names a spe-

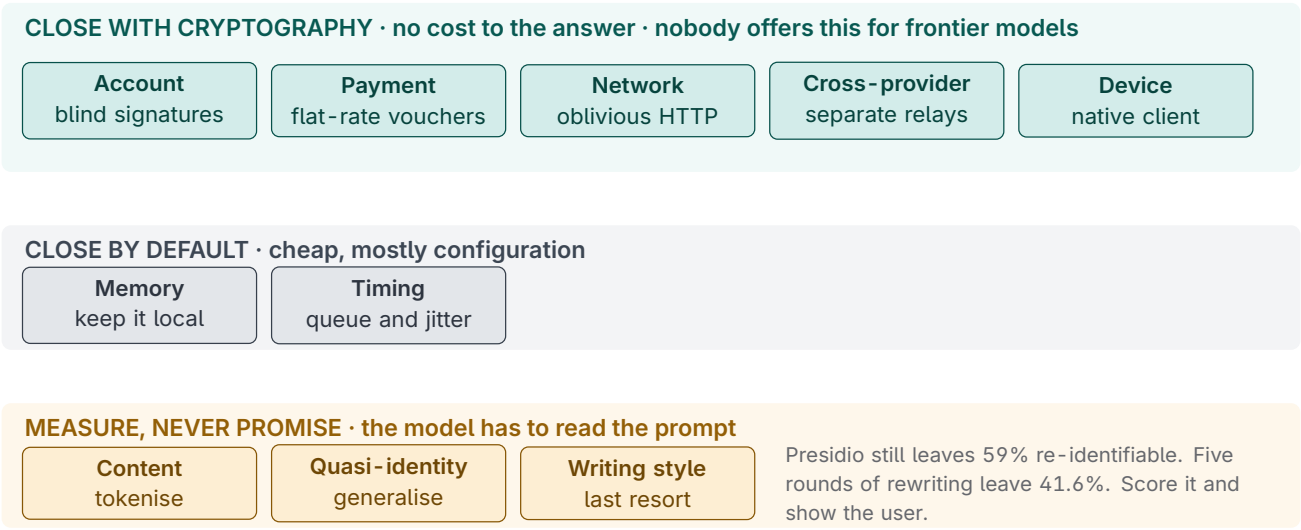


Figure 3. Ten channels build a profile. Five of them close with cryptography or configuration at no cost to answer quality, and nobody offers those for frontier models today. Three are partial by nature, because a hosted model must read what you write. We lead with the first group and we measure the last.

cific moment where one of you had real data, a model that would have answered it well, and no safe way to combine them. Name the company, the data and the workaround you built. If either of you has shipped security, privacy, infrastructure or inference systems, name the system and the scale. Specificity is the only signal here.]

What we can show without our own story. Practitioners keep building this for themselves: `DontFeedTheAI` has 655 stars, and Anonproxy is a cleaner reimplementation of it written by one person at a pentest firm who needed to send client data to Claude and would not. Nobody writes that much infrastructure for a problem they do not have, and nobody writes it at all if something is purchasable. A third of employees use personal accounts for AI work, and the share of enterprise AI traffic carrying sensitive data went from 10.7 % to 39.7 % in two years [9].

And the evidence cuts the other way too, which we would rather state than have discovered. Brave converts 101 million privacy-selected users into roughly a hundred thousand VPN subscribers, a tenth of one percent. Mozilla earns $66.4M from subscriptions and advertising combined across about 200 million users. The counter-example that matters is Venice.ai at roughly $70M of annual recurring revenue, profitable, on an explicit no-logging pitch: paid AI privacy rides the twenty-dollar AI subscription anchor, not the five-dollar privacy anchor.

For institutions the evidence is the restrictions themselves: JPMorgan, Goldman Sachs, Citi, Deutsche Bank, Bank of America and Wells Fargo all limited frontier models over confidentiality [15], and the rest run weaker models in-house to keep their name off the request. They are giving up capability to hide intent, which is exactly the trade a pool removes.

> HOW WE WOULD KNOW WE ARE WRONG
> If ten funds will not tell us what they would pay to use frontier models for deal work without being identified, or three will not join a founding cohort, the institutional thesis is wrong and no consumer growth rescues it.

06 Competitors, and what we understand

Who are your competitors? What do you understand that they don't?

Reversible tokenisation, vendor-side. Protecto, WitnessAI, Skyflow, Kong's sanitizer with its restore flag, Tonic. The vendor holds the keys and none has an agent path. Skyflow now ships a de-identify and re-identify gateway.

Reversible tokenisation, open source. Anonproxy is the best-engineered at 4,750 lines and 120 tests, and has 7 stars. Presidio has 10,931 and left Microsoft for a neutral foundation. Dataiku ships a fully reversible proxy as Apache-2.0 with no paid tier. LLM Guard, the most-cited comparison, was archived in July 2026. Detection itself is free from OpenAI and NVIDIA.

Confidential compute. Tinfoil, Confer, Privatemode, Apple PCC. A stronger promise on the content channel, limited to open weights, and positioned explicitly against redaction. That objection is coming and it is answered in §4 rather than avoided.

Finance AI and anonymous access. AlphaSense, Hebbia and Rogo sell answers to the same firms; Rogo was valued at $2B in April [17]. 1Privacy.ai, nullsink and Venice.ai sell private access to individuals. None hides the firm asking, and we found nobody selling anonymous frontier-model access to institutions as a shared pool.

What we understand that they do not. Four things.

1. **The surrogate was never the linkage channel.** Everyone selling reversible tokenisation implies it prevents profiling. Carlini and Tramèr's group link the same person's pseudonymous profiles at up to 68 % recall at 90 % precision, against near zero for the best non-LLM method, including across a deliberately removed one-year gap [10]. Topic, expertise and recurring specifics carry identity. Rotating a fake name buys nothing, and we say so in public.
2. **The cheap channels are the ones nobody closed.** The industry spent three years on the expensive, partial channel and skipped the four that close with cryptography at no cost.
3. **Identity is a negative claim, and negative claims need a third party.** This is the structural argument in §4 and it is

the only durable answer to platform risk.

4. **For an institution, the secret is who is asking.** Everyone else hides what is in the prompt. A fund's question about a company is valuable precisely because of whose question it is, and a single customer cannot hide itself; only a crowd can.

07 Technical stack

What tech stack are you using? Include AI models and AI coding tools.

Rust for the client and the daemon: the token store, the Oblivious HTTP encapsulation, the streaming path and the padding scheduler, compiled to one signed binary per platform. Python for the detection and evaluation work, where the model ecosystem lives.

Cryptography from published specifications rather than invention. Blind RSA per RFC 9474 and VOPRF per RFC 9497 for tokens, in the Privacy Pass framing of RFC 9576 through 9578. Oblivious HTTP per RFC 9458, with HPKE for encapsulation, against commercially operated relays from Fastly or Cloudflare, chosen per provider so the relay is never the provider's own edge. Token keys wrapped by the Apple Secure Enclave and by TPM 2.0 elsewhere.

On-device scoring uses an encoder tier through ONNX Runtime with CoreML on Apple silicon, GLiNER2-PII and OpenAI's Apache-2.0 Privacy Filter, with a small local model, Qwen3 at 0.6 to 1.7 billion parameters on MLX or llama.cpp, for the contextual pass. Nothing in the hot path calls a cloud model. Surrogates derive by HMAC-SHA256 over a scope key from HKDF; the vault is SQLite with SQLCipher.

Evaluation runs a paired harness: the same task with and without the harness, across three frontier models, judged blind to condition, with exact match where the answer is structured.

AI coding tools. Claude Code is the primary agent and writes the first draft of most things, with Codex CLI as an adversarial second reader on security-sensitive diffs. Claude Opus and Fable for architecture and review. We dogfood: our own agents reach those models through Lethe, so our repository and our own identities travel the path we are shipping, which has been the best bug-finder we have.

08 Business model

How will you make money? How much could you make?

Institutions already pay to hide what they are about to do. Roughly 15 to 18 % of US equity volume trades in dark pools, and about 44 % off-exchange in total, so a large order does not move the market before it fills [16]. Lethe is a dark pool for AI. The firms that pay are the ones whose questions are worth more than their answers.

Three revenue lines. **Firm membership**, $100,000 to $500,000 a year by size, is the profit centre, and includes the firm's own compliance ledger, policy controls and single sign-on. Analysts already pay $10,000 to $20,000 a seat for AI research tools [17] and hedge funds average $1.6M a year on alternative data [18]. **A per-query fee** sits on usage passed through at cost, the way a dark pool charges per share, so heavy users never sink the margin. **The consumer app**, about $20 a month, is priced near break-even; its job is to make the pool large, because a large pool is what firms pay for.

Compliance is a selling point. US regulators have fined financial firms more than $2 billion since 2021 for business communications they failed to record [19]. Anonymous AI with no record would be a liability. Lethe writes every request to a ledger the firm holds: the compliance team keeps everything; the provider keeps nothing that points to the firm.

Lethe holds the provider contracts. Members are anonymous behind it. The pitch to providers is net-new demand: the work in the pool is work they do not get today, because its owners restrict frontier models or run weaker ones in-house.

How big, honestly. Illustrative, not a forecast: 500 member firms at $250,000 a year is $125M of annual revenue before usage fees; 2,000 is $500M. That is a different company from the security-tool path, where the median contract is about $25,000 and the closest exit sold for $117M. The constraint is not price, it is the crowd: a pool with one fund in it hides nobody. This is a network-effects business, the hardest kind to start and the hardest kind to catch.

09 What we tested and discarded

This proposal replaces an earlier one. Recording why is part of the argument.

The earlier plan was reversible identity tokenisation with per-provider surrogates. Six findings ended it. The mechanism was already published in June 2026, including the quasi-identifier scoring we had claimed as a gap [11]. Per-provider rotation does not defeat linkage [10]. Entity replacement leaves 59 % re-identifiable under Presidio and 41.6 % after five rounds of adversarial rewriting, against a regulatory threshold of 5 % [12]. The mechanism is patented by a licensing company, granted in September 2025 with live continuations [13]. The consumer redaction thesis has a graveyard: Crosshatch, Blyss, Piiano's vault. And the analogy we were using, Datavant, is backwards, because Datavant holds the master salt and holding it is the business.

> **ONE ASSET FELL OUT OF IT**
> The paper everyone cites for "anonymisation makes models dumber" [20] measured bracketed tags rather than realistic surrogates, and its own worked example redacts the words *Earth* and *Sun*. Nobody has run that experiment properly. Doing so across three frontier models is a few weeks, is publishable, and would let us own the evaluation frame for the category.

10 Risks, and what we do not claim

Adversary	Residual
Provider identifying the account	● none, by construction
Provider linking payment to use	● none, by construction
Edge network joining providers	● relay diversity required
Provider memory features	● we do not use them
Traffic analysis on the stream	◑ padding closes most of it
Content and quasi-identifiers	○ measured, reported, never zero
Writing style across sessions	○ partial, expensive, optional
Small pool at launch	○ founding cohort, plus consumers
Unique question, one asker	◑ hides who, not what

● closed ◑ partial ○ open

Three constraints decide whether this is a company. The first is the cold start: a pool starts empty, and dark pools solved this by launching with a founding consortium of buyers, which is our plan for firms, with the consumer app filling the crowd from day one. The second is provider approval for institutional access, which is closer to reselling than an app is. The third is accountability, not resale. Anthropic's commercial terms explicitly permit using the API "to power products and services Customer makes available to its own customers and end users" (A.1), and restrict only reselling the Services themselves without approval (D.4). An app serving our own subscribers is permitted; anonymous raw API access sold to developers would need approval. What remains is that we answer for our users' compliance, and anonymity makes abuse prevention our problem. The answer is the one Apple, Meta and Kagi use: credentials rate-limited by construction, each good for a finite number of requests. OpenAI's and Google's terms still need the same read. Throughout, the anonymity set is the membership: a blind signature is only as anonymous as the crowd holding one, which makes the privacy story and the growth story the same story.

11 What we would file

Drafted narrowly against Datavant and Anonos. First, provider-scoped surrogate derivation where the salt binds to the recipient model provider's identity, the purpose is preventing rather than enabling linkage, and resolution is device-local. Second, negative-disclosure attestation: a hash-chained signed ledger committing to the set of entities withheld at a boundary, which inverts the usual transparency log and appears unclaimed. Third, detokenisation bound to a tool's action boundary. We would not file on PII detection, surrogate substitution as such, or vault round trips; all are owned or pre-2020 prior art. Before committing engineering to any tokenisation architecture, counsel answers one question: whether the AI-inference limitations in the Anonos grant are supported by its 2013 provisional or only by the 2021 and 2022 continuations.

12 The name

Lethe: the river in Hades whose water erases memory. Mnemosyne: the pool that restores it. Orphic initiates were told to avoid the first and drink from the second. The provider drinks from Lethe and never learns who was asking. You, and your compliance team, keep Mnemosyne.

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